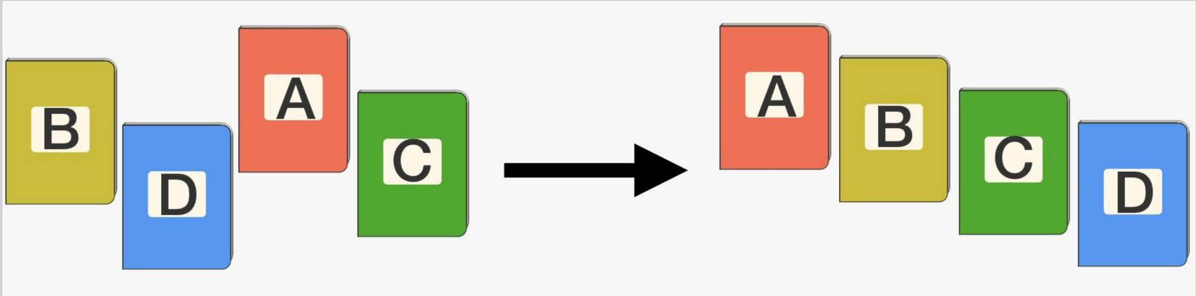
Counting Sort

MODULE 3

Sorting

- Sorting is arranging the elements in ascending and descending order



Procedure to perform counting sort

- Let us an Array A of N elements is in memory.

Step1:

- Find out the maximum element from the given array.

max		0	1	2	3	4	5	6
8		4	2	2	8	3	3	1

Step2:

- Initialize an array of length max+1 with all the elements 0. This array is used for storing the count of the elements in the array.

	0	1	2	3	4	5	6	7	8
Count array	0	0	0	0	0	0	0	0	0

Procedure to perform counting sort

- Let us an Array A of N elements is in memory.

Step3:

- Store the count of each element of Input array at their respective index in count array.

Input Array

0	1	2	3	4	5	6
4	**2**	**2**	**8**	**3**	**3**	**1**

Count array

0	1	2	3	4	5	6	7	8
0	**1**	**2**	**2**	**1**	**0**	**0**	**0**	**1**

Procedure to perform counting sort

- Let us an Array A of N elements is in memory.

Step4:

- Store cumulative sum of the elements of the count array. It helps in placing the elements into the correct index of the sorted array.

	0	1	2	3	4	5	6
Input Array	**4**	**2**	**2**	**8**	**3**	**3**	**1**

	0	1	2	3	4	5	6	7	8
Count array	**0**	**1**	**2**	**2**	**1**	**0**	**0**	**0**	**1**

Count array Cumulative sum	**0**	**1**	**3**	**5**	**6**	**6**	**6**	**6**	**7**
	0	1	2	3	4	5	6	7	8

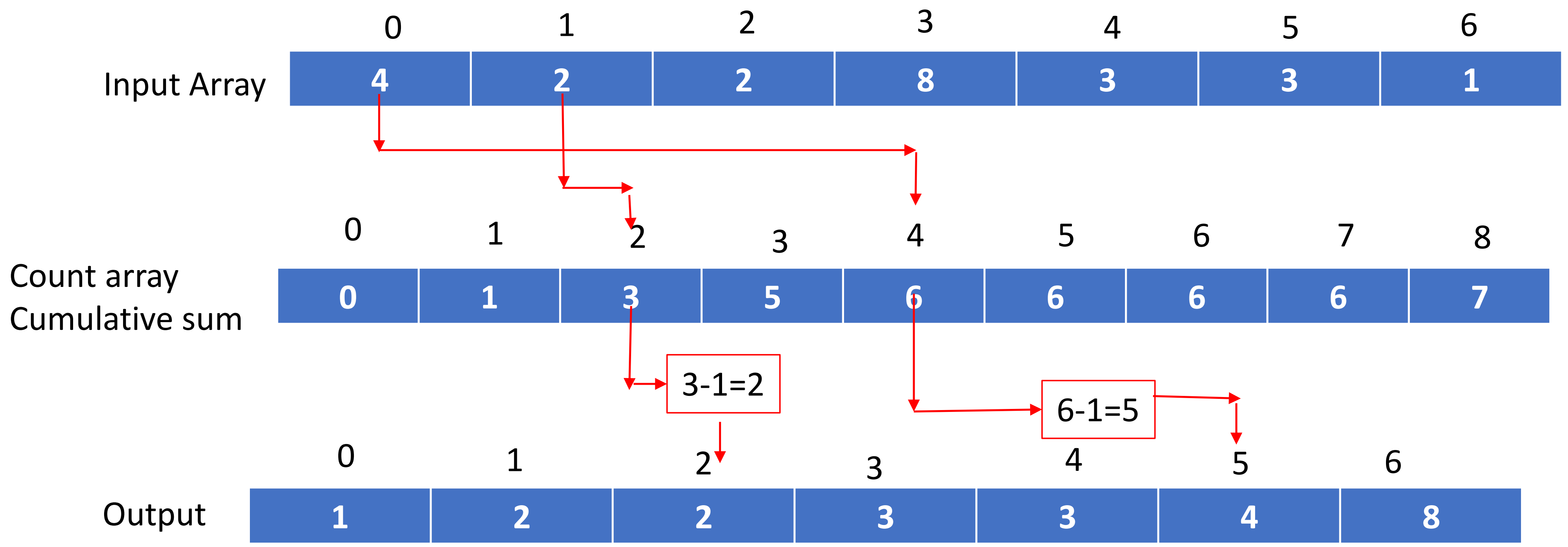
Cumulative count

Procedure to perform counting sort

- Let us an Array A of N elements is in memory.

Step5:

- Find the index of each element of the original array in the count array. This gives the cumulative count. Place the element at the Index calculated as shown below.



Step6: After placing each element at its correct position, decrease its count by one

Procedure to perform counting sort

Step6:

After placing each element at its correct position, decrease its count by one

Output Array

0	1	2	3	4	5	6
1	2	2	3	3	4	8

Input Array

0	1	2	3	4	5	6
4	2	2	8	3	3	1

```
CountingSort(array, size)
  max <- find largest element in array
  initialize count array with all zeros
  for j <- 0 to size
    find the total count of each unique element and
    store the count at jth index in count array
  for i <- 1 to max
    find the cumulative sum and store it in count array itself
  for j <- size down to 1
    restore the elements to array
    decrease count of each element restored by 1
```

End